

Digital Electronics and Logic Design (ELE1204)

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Chapter 2

- **Boolean Algebra**
 - Truth tables
- **Logic Simplification**
 - Karnaugh Maps (2,3,4 variables)
- **Design using Simplified Expressions**
 - Realization of Boolean functions using logic gates
 - Introduction to Digital circuit Simulation Tools (Logisim, multisim)

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Truth table Examples

AND gate

Expression: $X = A \cdot B$

A	B	X
0	0	0
1	0	0
0	1	0
1	1	1

OR gate

Expression: $X = A + B$

A	B	X
0	0	0
1	0	1
0	1	1
1	1	1

NOT gate

Express $X = \overline{X}$

A	X
0	1
1	0

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Truth table Examples

XOR: $X = A \oplus B$

A	B	X
0	0	0
1	0	1
0	1	1
1	1	0

Quick Exercise: Draw the truth table for **XNOR**

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Duality principle

Original: $A + (B \cdot C) = (A + B) \cdot (A + C)$

Dual: $A \cdot (B + C) = (A \cdot B) + (A \cdot C)$

Exercise: Prove these using Truth tables

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- Introduction to Logic Simplification

Logic simplification is the process of reducing a Boolean expression or logic circuit to its simplest form. This helps in:

- Minimizing the number of **logic gates** used in a circuit.
- Reducing **power consumption** and **cost**.
- Improving **circuit speed** and **efficiency**.

Methods of Logic Simplification

1. **Boolean Algebra** – Using Boolean laws and theorems to simplify expressions.
2. **Karnaugh Maps (K-Maps)** – A visual method for minimizing Boolean functions.
3. **Quine-McCluskey Method** – A tabular method used for complex expressions (out of scope).

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- **Karnaugh Maps (K-Maps)**

A **K-Map** is a graphical tool used to simplify Boolean expressions by organizing truth table values into a grid format.

Key Rules for K-Maps

1. Each cell represents a minterm (SOP) or maxterm (POS).
2. Adjacent cells differ by only one variable.
3. Grouping adjacent 1s minimizes the expression.
4. Groups must be in powers of 2 (1, 2, 4, 8, etc.).
5. Wrap-around is allowed (K-Maps are cyclic).

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- Karnaugh Maps for 2, 3, and 4 Variables

2 - Variable K-Map

A **2-variable K-Map** has **4 cells** since $2^2 = 4$

		B	
		0	1
A	0	m0	m1
	1	m2	m3

$$m0 = A'B' \quad m1 = A'B$$

$$m2 = AB' \quad m3 = AB$$

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- Karnaugh Maps for 2, 3, and 4 Variables

2-Variable K-Map

$$F_1 = \Sigma(0, 1, 3)$$

A	B	minterms	F1	m
0	0	$\overline{A}\overline{B}$	1	m0
0	1	$\overline{A}B$	1	m1
1	0	$A\overline{B}$	0	m2
1	1	AB	1	m3

$$F_1 = \overline{A}\overline{B} + \overline{A}B + AB$$

$$F_1 = \Sigma(0, 3)$$

		B	
		0	1
A	0	1	1
	1	0	1

$$F_1 = \overline{A}\overline{B} + \overline{A}B + AB$$

$$F_1 = \overline{A}(\overline{B} + B) + B(\overline{A} + A)$$

$$F_1 = \overline{A} + B$$

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- Karnaugh Maps for 2, 3, and 4 Variables

3-Variable K-Map

A	B	C	M
0	0	0	m0
0	0	1	m1
0	1	0	m2
0	1	1	m3
1	0	0	m4
1	0	1	m5
1	1	0	m6
1	1	1	m7

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- **Karnaugh Maps for 2, 3, and 4 Variables**

3-Variable K-Map Exercise

3.1* Simplify the following Boolean functions, using three-variable maps:

(a) $F(x, y, z) = \Sigma(0, 2, 4, 5)$

(b) $F(x, y, z) = \Sigma(0, 2, 4, 5, 6)$

(c) $F(x, y, z) = \Sigma(0, 1, 2, 3, 5)$

(d) $F(x, y, z) = \Sigma(1, 2, 3, 7)$

3.2 Simplify the following Boolean functions, using three-variable maps:

(a)* $F(x, y, z) = \Sigma(0, 1, 5, 7)$

(b)* $F(x, y, z) = \Sigma(1, 2, 3, 6, 7)$

(c) $F(x, y, z) = \Sigma(2, 3, 4, 5)$

(d) $F(x, y, z) = \Sigma(1, 2, 3, 5, 6, 7)$

(e) $F(x, y, z) = \Sigma(0, 2, 4, 6)$

(f) $F(x, y, z) = \Sigma(3, 4, 5, 6, 7)$

3.3* Simplify the following Boolean expressions, using three-variable maps:

(a)* $xy + x'y'z' + x'yz'$

(b)* $x'y' + yz + x'yz'$

(c)* $F(x, y, z) = x'y + yz' + y'z'$

(d) $F(x, y, z) = x'yz + xy'z' + xy'z$

Further reading - K-map tutorial: <https://www.youtube.com/watch?v=lw1STgKUpW0>

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- **Realization of Boolean functions using logic gates**

The realization of **Boolean functions** using logic gates involves designing and implementing digital circuits that perform logical operations based on Boolean algebra. **Logic gates** such as AND, OR, NOT, NAND, NOR, XOR, and XNOR are used to construct these circuits.

Why is Boolean Function Realization Important?

- ✓ Helps in **minimizing circuits** for cost and power efficiency.
- ✓ Converts **Boolean expressions into physical logic circuits**.
- ✓ Essential in **digital circuit design**, including ALUs, memory units, and control systems.

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Steps to Realize a Boolean Function Using Logic Gates

1. **Obtain the Boolean Expression:**
 - The Boolean function may be given in **Sum of Products (SOP)** or **Product of Sums (POS)** form.
2. **Simplify the Expression** (if necessary):
 - Use **Boolean algebra** or **Karnaugh Maps (K-Maps)** to reduce the number of terms and gates.
3. **Implement Using Basic Logic Gates:**
 - Use **AND, OR, and NOT** gates for **SOP implementation**.
 - Use **NOR and NAND** gates for **POS and universal gate realization**.
4. **Draw the Logic Circuit:**
 - Connect logic gates according to the Boolean expression.
 - Verify correctness using a **truth table** or **simulation tools** like **Logisim** or **Multisim**.

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Example 1: SOP Realization

Boolean Function: $F(A,B,C) = AB + A'C$

Step 1: Identify Gates Needed

- AB requires an **AND gate**.
- A' requires a **NOT gate**.
- $A'C$ requires another **AND gate**.
- The final sum $AB + A'C$ requires an **OR gate**.

Step 2: Draw the Circuit

- **AND gate** for AB .
- **NOT gate** for A' .
- **AND gate** for $A'C$.
- **OR gate** to combine both terms.

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Truth table

$$F(A,B,C) = AB + A'C$$

A	B	C	AB	A'	$A'C$	F
0	0	0	0	1	0	0
0	0	1	0	1	1	1
0	1	0	0	1	0	0
0	1	1	0	1	1	1
1	0	0	0	0	0	0
1	0	1	0	0	0	0
1	1	0	1	0	0	1
1	1	1	1	0	0	1

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- **Reading and Practice assignment**
 - Further reading: Read Chapter 3 - Digital Design 5th Edition by Morris Mano

Thank you